

Effect Modification / Interaction

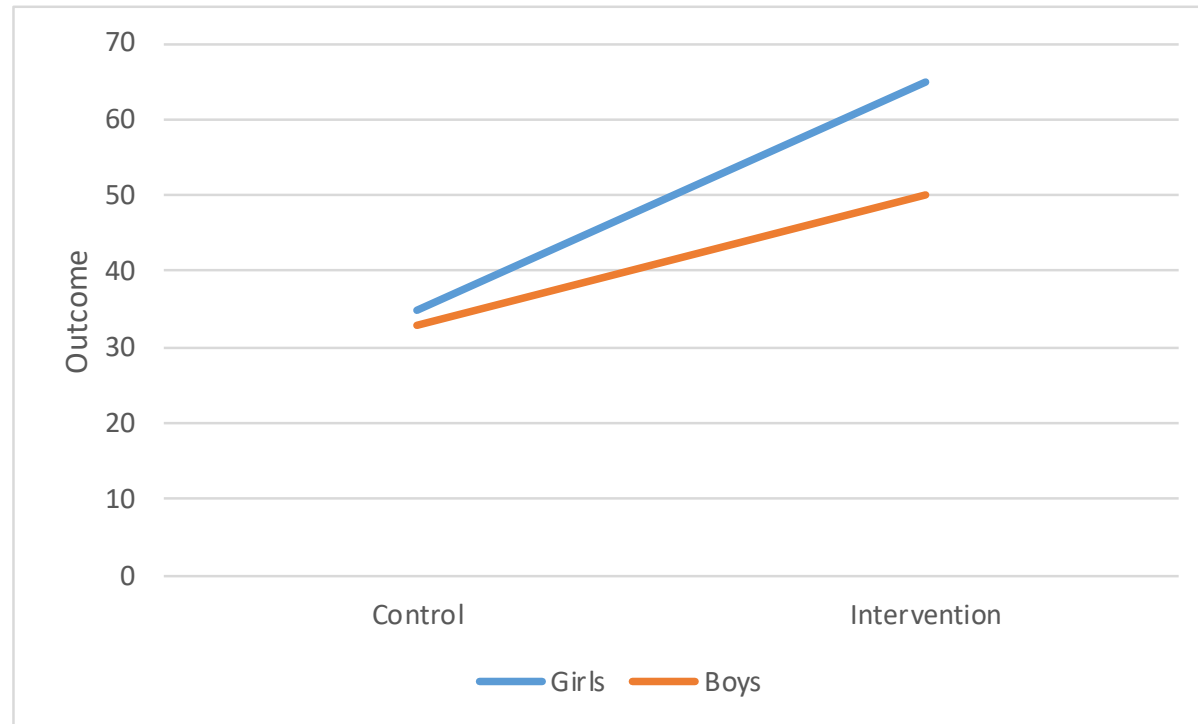
PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Effect Modification

- The effect of an intervention differs depending on another factor. For example:
 - An intervention has a bigger effect on an outcome in girls compared with boys
 - A school-based intervention is only effective for children who receive good homework support at home



Adding Interaction

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, b \\ k = 1, 2, \dots, n_{ij} \end{cases}$$

$$\varepsilon_{ijk} \sim iid N(0, \sigma^2)$$

- τ_i is effect of treatment i (ignoring effect modifier)
- β_j is effect of effect modifier j (ignoring treatment)
- $(\tau\beta)_{ij}$ is the interaction effect of treatment i and effect modifier j
 - This is the deviation from an additive model *Multiplicative Effect*

ANOVA Table

Source of variation	Sum of squares	Degrees of freedom	Mean square	F
Treatment	SS_{τ}	$a-1$	$MSA = SS_{\tau}/(a-1)$	MSA/MSE
Effect modifier	SS_{β}	$b-1$	$MSB = SS_{\beta}/(b-1)$	MSB/MSE
Treatment x effect modifier interaction	$SS_{\tau\beta}$	Multiplicative Effect $(a-1)(b-1)$	$MSAB = SS_{\tau\beta}/((a-1)(b-1))$	$MSAB/MSE$
Error	SS_E	$\sum\sum n_{ij} - ab$	$MSE = SS_E/(\sum\sum n_{ij} - ab)$	
Total	SS_T	$\sum\sum n_{ij} - 1$		

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